

2. Functional Dependencies & Minimal Set

User	User_id → full_name
	User_id → email
	User_id → phone
	User_id → password
	User_id → gender
	User_id → created_at
	User_id → statu
Host	user_id → host_since
	user_id → host_rating
	user_id → verification_status
Guest	user_id → loyalty_points
	user_id → travel_preferences
Admin	user_id → role
	user_id → access_level
Property	property_id → host_id
	property_id → title
	property_id → description
	property_id → property_type
	property_id → country
	property_id → city
	property_id → state
	property_id → pincode
	property_id → latitude
	property_id → longitude
Listing	listing_id → property_id
	listing_id → created_at
	listing_id → availability_status
	listing_id → price_per_night
	listing_id → max_guests
	listing_id → bedrooms
	listing_id → bathrooms
	listing_id → house_rules

Amenity	amenity_id → amenity_name
	amenity_id → description
Listing_Amenity	(listing_id, amenity_id) → (no non-key attributes)
Booking	booking_id → guest_id
	booking_id → listing_id
	booking_id → check_in_date
	booking_id → check_out_date
	booking_id → booking_date
	booking_id → no_of_guests
	booking_id → booking_status
	booking_id → total_amount
	booking_id → admin_id
Payment	payment_id → booking_id
	payment_id → amount
	payment_id → payment_date
	payment_id → payment_method
	payment_id → payment_status
	payment_id → transaction_ref
Review	review_id → booking_id
	review_id → guest_id
	review_id → listing_id
	review_id → rating
	review_id → comment
	review_id → review_date
Cancellation	cancellation_id → booking_id
	cancellation_id → cancelled_by
	cancellation_id → cancellation_date
	cancellation_id → refund_amount
	cancellation_id → reason

3. BCNF Check:

- **User**

Candidate Keys	{ user_id }
BCNF Check	For every FD $X \rightarrow Y$: $X = \text{user_id}$ which is a superkey. No partial or transitive dependencies exist.
Verdict	IN BCNF. No decomposition needed.

- **Host**

Candidate Keys	{ user_id }
BCNF Check	user_id is the PK and superkey. Every non-trivial FD has the superkey on LHS.
Verdict	IN BCNF. No decomposition needed.

- **Guest**

Candidate Keys	{ user_id }
BCNF Check	user_id is the PK . The single FD has user_id on LHS, which is a superkey. No partial or transitive dependencies exist
Verdict	IN BCNF. No decomposition needed.

- **Admin**

Candidate Keys	{ user_id }
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BCNF Check	user_id is the PK. All FDs are determined by the superkey.
Verdict	IN BCNF. No decomposition needed.

- **Property**

Candidate Keys	{ property_id }
BCNF Check	property_id is the PK. All non-trivial FDs have property_id on the LHS, which is a superkey. No dependency among non-key attributes
Verdict	IN BCNF. No decomposition needed.

- **Listing**

Candidate Keys	{ listing_id }
BCNF Check	listing_id is the PK. Every FD is of the form listing_id → attribute. No non-key attribute determines another
Verdict	IN BCNF. No decomposition needed.

- **Amenity**

Candidate Keys	{ amenity_id }
BCNF Check	amenity_id is the PK. All attributes are functionally determined solely by amenity_id. No transitive or partial dependencies exist
Verdict	IN BCNF. No decomposition needed.

- **Listing_Amenity**

Candidate Keys	{ (listing_id, amenity_id) }
BCNF Check	The entire relation is the composite key. There are no non-key attributes and hence no non-trivial FDs that could violate BCNF.
Verdict	IN BCNF. No decomposition needed.

- **Booking**

Candidate Keys	{ booking_id }
BCNF Check	booking_id is the PK. All attributes are functionally determined solely by booking_id. No transitive or partial dependencies exist.
Verdict	IN BCNF. No decomposition needed.

- **Payment**

Candidate Keys	{ payment_id }
BCNF Check	payment_id is the PK. transaction_ref is unique but treated as a surrogate key attribute, not used as a separate determinant in FDs. All FDs have payment_id on LHS
Verdict	IN BCNF. No decomposition needed.

- **Review**

Candidate Keys	{ review_id }
BCNF Check	review_id is the PK. All non-trivial FDs have the superkey on LHS
Verdict	IN BCNF. No decomposition needed.

- **Cancellation**

Candidate Keys	{ cancellation_id }
BCNF Check	cancellation_id is the PK. All non-trivial FDs have the superkey on LHS
Verdict	IN BCNF. No decomposition needed.